

BONILLA ISLAND, BC, Canada

WMO: 714840

Lat: 53.493N

Lon: 130.637W

Elev: 41

StdP: 14.67

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.4	28.4	3.7	6.7	28.3	9.4	8.9	32.1	53.3	42.0	47.1	43.4	8.9	80	0.839

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.3	61.4	56.9	59.9	56.6	58.7	55.9	58.4	60.0	57.4	59.0	56.7	58.2	9.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.4	70.4	59.3	56.5	68.1	58.5	55.7	66.2	57.8	25.3	60.3	24.7	59.2	24.2	58.4	64.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 46.0	39.9	34.6	DB	20.4	66.8	6.2	4.5	15.9	70.0	12.3	72.7	8.8	75.2	4.3	78.4	(4)
(5)			WB	14.1	61.4	7.3	1.7	8.9	62.6	4.7	63.6	0.6	64.5	-4.7	65.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	41.3	40.8	41.5	44.7	48.3	52.1	55.1	56.2	54.3	49.6	44.6	41.3
	DBStd	6.73	5.65	4.41	4.40	3.09	2.72	2.37	2.20	2.00	2.22	2.92	4.76	5.11
	HDD50	1499	270	257	263	162	65	5	0	0	1	40	166	270
	HDD65	6371	734	676	727	609	517	387	306	272	321	476	611	735
	CDD50	602	1	0	1	4	13	68	159	193	130	28	5	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	15.7	17.5	16.5	16.8	16.2	15.1	14.1	13.2	12.9	14.1	17.1	17.4	17.7
Precipitation	PrecAvg	85.4	8.5	7.6	7.2	6.5	4.8	4.0	3.5	4.4	6.9	11.3	10.4	9.7
	PrecMax	103.1	12.5	15.2	13.3	13.0	9.1	7.0	7.9	10.5	13.0	18.6	18.0	14.1
	PrecMin	67.9	2.5	0.8	1.7	1.9	1.2	0.6	1.0	0.6	2.1	5.1	4.3	4.1
	PrecStd	10.0	2.6	3.5	2.9	2.9	2.2	1.7	2.1	2.3	2.6	3.0	3.5	2.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	52.2	54.1	57.9	59.4	62.3	62.6	63.3	63.2	57.5	58.9	52.3
		MCWB	44.7	44.1	44.7	48.0	49.2	56.3	58.1	59.1	54.7	51.0	54.6	46.4
	2%	DB	49.9	49.0	49.4	52.0	54.4	57.9	60.3	61.4	59.6	55.4	52.9	49.8
		MCWB	45.6	44.8	45.5	47.4	50.2	54.3	57.2	58.1	55.6	52.5	48.7	46.6
	5%	DB	48.2	47.4	47.7	49.5	52.6	56.2	59.0	60.1	58.2	54.2	51.3	48.4
		MCWB	46.0	44.5	45.0	46.5	49.5	53.2	56.4	57.3	55.3	51.5	48.4	45.9
	10%	DB	47.1	46.0	46.5	48.2	51.4	54.9	57.9	58.9	57.2	53.3	50.1	47.3
		MCWB	45.0	43.4	44.1	45.6	48.8	52.3	55.6	56.5	54.9	50.7	47.6	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	48.1	48.3	49.9	53.0	56.7	59.2	60.1	58.0	55.2	58.6	49.1
		MCDB	49.4	49.2	50.4	53.3	55.6	59.2	61.1	61.8	59.1	55.9	60.4	50.0
	2%	WB	47.3	46.7	47.0	48.5	51.1	54.9	57.8	58.7	57.0	53.7	50.9	47.6
		MCDB	48.3	48.2	49.1	50.8	53.2	56.8	59.5	60.3	58.2	54.9	52.0	48.9
	5%	WB	46.5	45.6	46.0	47.5	50.1	53.9	57.0	57.7	56.3	52.6	49.7	46.4
		MCDB	47.7	47.2	48.0	49.5	51.8	55.7	58.6	59.3	57.6	53.9	51.1	47.8
	10%	WB	45.6	44.6	45.0	46.4	49.3	52.9	56.2	57.0	55.5	51.5	48.5	45.2
		MCDB	47.0	46.3	47.0	48.3	51.0	54.6	57.8	58.6	57.1	53.0	50.2	46.8
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.5	5.5	5.0	4.4	4.1	4.0	4.3	4.2	4.4	5.3	5.5
		MCDBR	5.7	6.7	6.4	7.1	6.2	6.1	5.5	5.9	5.8	5.1	5.8	5.6
		MCWBR	4.4	5.2	4.3	4.6	4.3	4.0	3.8	4.3	3.9	4.3	5.4	5.0
	5% WB	MCDBR	4.7	5.8	5.5	6.4	5.5	5.4	5.1	5.4	4.7	4.3	5.5	5.3
		MCWBR	4.6	5.1	4.3	4.7	4.1	4.0	3.9	4.4	4.0	4.4	5.6	5.3
Clear-Sky Solar Irradiance	taub	0.311	0.319	0.344	0.380	0.376	0.375	0.369	0.370	0.363	0.345	0.325	0.304	
	taud	2.411	2.499	2.439	2.324	2.402	2.440	2.499	2.511	2.496	2.451	2.336	2.339	
	Ebn at Noon	212	252	267	269	277	278	278	271	258	236	199	187	
	Edh at Noon	17	22	28	36	35	34	31	30	27	23	18	16	
All-Sky Solar Radiation	RadAvg	178	445	804	1260	1586	1656	1562	1329	899	506	237	145	
	RadStd	22	70	89	68	135	117	162	102	92	62	27	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page